Lovely Professional University

Lovely School of Computer Science and Engineering

PROJECT REPORT

(Project Semester January-April 2025)

Demographic Analysis of Car Sales in India: An Interactive Dashboard Approach

Submitted by

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Registration No: 12305340

Section: K23EG

Programme: B. Tech - Computer Science and Engineering

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering Lovely Professional University, Phagwara

Transforming Education Transforming India

CERTIFICATE

This is to certify that Raja bearing Registration no. 12305340 has completed INT217 (Introduction to Data Management) project titled, “Demographic Analysis of Car Sales in India: An Interactive Dashboard Approach” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature

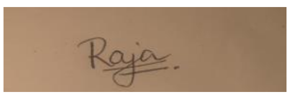
Jaffar Amin Chacket (UID: 30453)
Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University Phagwara, Punjab.

Date:

DECLARATION

I, Raja, student of B.Tech Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Raja 12305340- K23EG



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09/04/2025

Acknowledgement

The opportunity of attaining a course based on Data Management using Excel at Lovely Professional University was worth learning. It was a prestige for me to be part of it. During the period of my course, I received tremendous knowledge related to Microsoft Excel and Data Management.

Pre-eminently, I would like to express my deep gratitude and special thanks to my course teacher Jaffar Amin Chacket (UID: 30453)for his theoretical knowledge and encouragement on this project and for her valuable guidance and affection for the successful completion of this project.

Secondly, I would like to thank Lovely Professional University for giving me an opportunity to learn this course.

Lastly, I would like to thank the almighty and my parents for their constant encouragement, moral support, personal attention, and care.

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Abstract

Excel is a software program created by Microsoft that uses spreadsheets to organize numbers and data with formulas and functions. Excel analysis is ubiquitous around the world and used by businesses of all sizes to perform data analysis. Excel features calculation, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications, and several other features which make Excel a perfect choice to manage and analyse data.

My project is an Excel Dashboard. The Excel Dashboard is used to display overviews of large data tracks. Excel Dashboards use dashboard elements like tables, charts, and gauges to show the overviews. The dashboards ease the decision-making process by showing the vital parts of the data in the same window.

In this report, I have shared a project where I have done data analysis of a Suicide Rate dataset across US States. This report also presents my learning during my course classes.

Chapter 1 – Introduction

I have created an Excel Dashboard of a Suicide Rate dataset across US States. This dashboard explains and highlights important facts, trends, and patterns related to suicide rates in different states based on gender, race, age group, and year.

The dataset used contains information regarding suicide death rates in the United States. It includes details such as gender-wise data, race-wise data, Hispanic origin, and different age groups of people affected by suicide. The dataset also covers records from multiple years across various US states.

I have scrubbed and organized the entire dataset and performed the analysis on a clean data sheet created within the Excel workbook. I have derived and calculated important results from the dataset using various Excel features like pivot tables, slicers, and functions. I have represented the findings in the form of a dynamic dashboard using Excel's visualization tools and various charts.

Chapter 2 – Objectives

This project on Suicide Rate Analysis provides the records, facts, and trends of suicide rates across US States with respect to gender, race, age group, and year. However, here are the few main objectives that are discussed in the dashboard:

• Showing the total number of suicide cases and their distribution with respect to gender, race, age group, year, and state.
• Showing the performance and comparison of states overall, or in a particular year, based on gender-specific suicide rates.
• Showing the distribution and trends of suicide rates among different age groups.
• Highlighting the gender-wise comparison of suicide rates overall, or in specific states, or particular years.
• Showing the trend analysis of suicide rates over the years based on different states and demographic groups.
• Displaying the most affected age groups and races in terms of suicide rates.
• Highlighting the state-wise contribution to the total number of suicide cases.
• Visualizing the overall trends and patterns to provide meaningful insights for decision-making and awareness.

Chapter 3 - Source of Dataset

The dataset is taken from Data.gov. Data.gov is the official open data portal of the United States Government that provides access to a wide range of datasets for public use. This platform allows users to find and download datasets published by government agencies.

I have selected a Suicide Rate dataset which contains important details of suicide rates across the United States categorized by gender, race, age group, and year.

Here are the details of my chosen dataset:

Name - Death Rates for Suicide by Sex, Race, Hispanic Origin, and Age — United States
Link - https://catalog.data.gov/dataset/death-rates-for-suicide-by-sex-race-hispanic-origin-and-age-united-states-020c1
Format - CSV

Data Fields:

• INDICATOR — (String) — Description of the indicator measured

• UNIT — (String) — Unit of measurement (e.g., Rate per 100,000 population)

• UNIT_NUM — (Numeric) — Numerical code for unit type

• STUB_NAME — (String) — Classification by sex and race

• STUB_NAME_NUM — (Numeric) — Numerical code for STUB_NAME

• STUB_LABEL — (String) — Label for STUB_NAME category

• STUB_LABEL_NUM — (Numeric) — Numerical code for STUB_LABEL

• YEAR — (Integer) — Year of data record

• YEAR_NUM — (Numeric) — Numerical representation of year

• AGE — (String) — Age group category

• AGE_NUM — (Numeric) — Numerical representation of age group

• ESTIMATE — (Numeric) — Death rate estimate per specified population

Chapter 4 - Dataset Preprocessing

To ensure data quality and accuracy in analysis, preprocessing was carried out in the following steps:

1. Data Cleaning:
   - Removed unnecessary columns like timestamps and duplicate identifiers.
   - Standardized column headers for readability (e.g., `sellingprice`, `mmr`, `condition`).
   - Filled or removed missing entries in critical columns such as selling price and company.

2. Transformation:
   - Created new columns like `salesmonth` by combining year and month.
   - Derived averages and condition levels to prepare data for aggregation.

3. Structuring:
   - A clean dataset sheet was created with all filtered and transformed data.
   - The clean dataset was then used for building pivot tables and interactive visualizations in the dashboard.

Sample of Cleaned Dataset

year	company	model	trim	body	transmissi	vin	state	Grade	conditio	odomete	color	interior	seller	mmr	sellingpric	day	month	date	year2	Column2	Column3	salesmonth
2015	Kia	Sorento	LX	SUV	[illegible]	ca69fg5	PB	Poor	5	16639.0	white	[illegible]	otors ame	20500	21500	Tue	Dec	16	2014	[illegible]	-0800 (	Dec-2014
2015	Kia	Sorento	LX	SUV	[illegible]	ca69fg5	PB	Poor	5	9393.0	white	[illegible]	otors ame	20800	21500	Tue	Dec	16	2014	[illegible]	-0800 (	Dec-2014
2014	BMW	3 Series	328i SULE	Sedan	[illegible]	1c51ek1	PB	Excellent	45	1331.0	[illegible]	[illegible]	ices rema	31900	30000	Thu	Jan	15	2015	[illegible]	-0800 (	Jan-2015
2015	Volvo	S60	T5	Sedan	[illegible]	2tb4f13	PB	Excellent	41	14282.0	white	[illegible]	a rep/wo	27500	27750	Thu	Jan	29	2015	[illegible]	-0800 (	Jan-2015
2014	[illegible]	[illegible]	650i	Sedan	[illegible]	2c57ed1	PB	Excellent	43	2641.0	[illegible]	[illegible]	ices rema	66000	67000	Thu	Dec	18	2014	[illegible]	-0800 (	Dec-2014

Figure 2: Sample of Cleaned Dataset

Sample of Raw Dataset

year	company	model	trim	body	transmissi	vin	state	Grade	conditio	odomete	color	interior	seller	mmr	sellingpric	day	month	date	year2	Column2	Column3	salesmonth
2015	Kia	Sorento	LX	SUV	[illegible]	ca69fg5	PB	Poor	5	16639.0	white	[illegible]	otors ame	20500	21500	Tue	Dec	16	2014	[illegible]	-0800 (	Dec-2014
2015	Kia	Sorento	LX	SUV	[illegible]	ca69fg5	PB	Poor	5	9393.0	white	[illegible]	otors ame	20800	21500	Tue	Dec	16	2014	[illegible]	-0800 (	Dec-2014
2014	BMW	3 Series	328i SULE	Sedan	[illegible]	1c51ek1	PB	Excellent	45	1331.0	[illegible]	[illegible]	ices rema	31900	30000	Thu	Jan	15	2015	[illegible]	-0800 (	Jan-2015
2015	Volvo	S60	T5	Sedan	[illegible]	2tb4f13	PB	Excellent	41	14282.0	white	[illegible]	a rep/wo	27500	27750	Thu	Jan	29	2015	[illegible]	-0800 (	Jan-2015
2014	[illegible]	[illegible]	650i	Sedan	[illegible]	2c57ed1	PB	Excellent	43	2641.0	[illegible]	[illegible]	ices rema	66000	67000	Thu	Dec	18	2014	[illegible]	-0800 (	Dec-2014

Figure 1: Sample of Raw Dataset

Chapter 5 - Analysis on Dataset

General Description:

The dataset provides detailed information about suicide death rates across the United States. It covers various aspects and demographic details like gender, race, Hispanic origin, age group, and year-wise data. This comprehensive dataset allows for the observation of patterns and analysis of trends in suicide rates over different time periods and across various states.

This project focuses on analysing how suicide rates vary between different groups of people. The data helps in identifying which states have higher suicide rates, the comparison between male and female suicide rates, the most vulnerable age groups, and the differences in suicide rates among different races and ethnic groups.

The dataset also enables the study of the changing patterns of suicide rates over the years, highlighting any increase or decrease based on several factors. The clean dataset sheet was created to ensure that the data was accurate, consistent, and structured properly for effective analysis. Various Excel tools like Pivot Tables, Charts, Slicers, and Conditional Formatting were used to generate insights and represent them visually through a dynamic dashboard.

This analysis provides meaningful information that can help researchers, policymakers, and healthcare planners understand the suicide trends in the United States and make data-driven decisions to address this serious issue. The clean dataset was used to generate insights and create various visualizations for the dashboard.

• **Specific Requirements:**

The specific requirements of this project were to analyse the suicide rate dataset and provide meaningful insights using Excel. The project aimed to use the capabilities of Excel to turn raw data into understandable and visually engaging insights through a dashboard. The analysis and dashboard creation were focused on enabling easier decision-making and presenting key data in an effective way.

The main requirements included:

• Cleaning and transforming the raw dataset for accurate and meaningful analysis.
• Creating a separate clean dataset sheet within the Excel workbook for structured analysis and easy accessibility.

• Generating pivot tables to analyse suicide rates state-wise, gender-wise, race-wise, and age-wise.
• Designing various charts like bar charts, line charts, pie charts, and heatmaps to visualize different aspects of the data.
• Incorporating slicers to filter the dataset interactively based on parameters like year, state, gender, and age group.
• Highlighting key performance indicators (KPIs) like states with the highest suicide rates, lowest suicide rates, and average suicide rates across the dataset.
• Ensuring the dashboard is dynamic, user-friendly, and provides a comprehensive overview of the data.
• Utilizing conditional formatting to visually emphasize significant values and trends in the data.

These requirements were fulfilled using Excel tools and functions, ensuring an effective analysis of the dataset and the creation of a dynamic and insightful dashboard that can help users better understand suicide trends in the United States and facilitate data-driven decision-making.

• **Analysis Results:**

Based on the analysis of the suicide rate dataset using Excel, the following key results and insights were obtained:

• Gender-wise analysis revealed that the suicide rates were generally higher among males compared to females across most states and years. This insight highlights the importance of creating gender-specific mental health programs to address the issues faced predominantly by the male population.

• Age-wise analysis highlighted that certain age groups were more vulnerable to suicide, particularly middle-aged and elderly populations in some regions. The analysis also provided insights into the need for special attention and mental health support for younger populations to prevent future increases in suicide rates.

• Race and Hispanic origin analysis helped identify disparities in suicide rates among different racial groups, emphasizing the need for targeted mental health programs that cater to the unique cultural and social challenges faced by each group.

• Year-wise trend analysis showed variations in suicide rates over time, providing valuable insights into whether the rates increased, decreased, or remained stable in specific states or demographic categories. These trends help understand the effectiveness of past mental health initiatives and provide a base for planning future programs.

• The use of slicers and KPIs in the dashboard allowed users to interactively filter the data and focus on specific areas of interest, enhancing the overall analytical experience. The interactivity of the dashboard enables users to make quick and informed decisions by analysing different segments of the dataset.

These results provide a clear understanding of suicide rate patterns across the United States and support data-driven decision-making for awareness campaigns and preventive measures. The analysis also showcases the significance of using Excel's powerful features for handling large datasets and converting them into useful insights for practical applications.

• **Visualization:**

This section presents the visualizations created during the analysis of the Suicide Rate dataset using Excel. These visualizations provide a clear and interactive representation of the data for better understanding and decision-making.

The following visualizations were created:

Figure 1: Top Companies by Average Selling Price

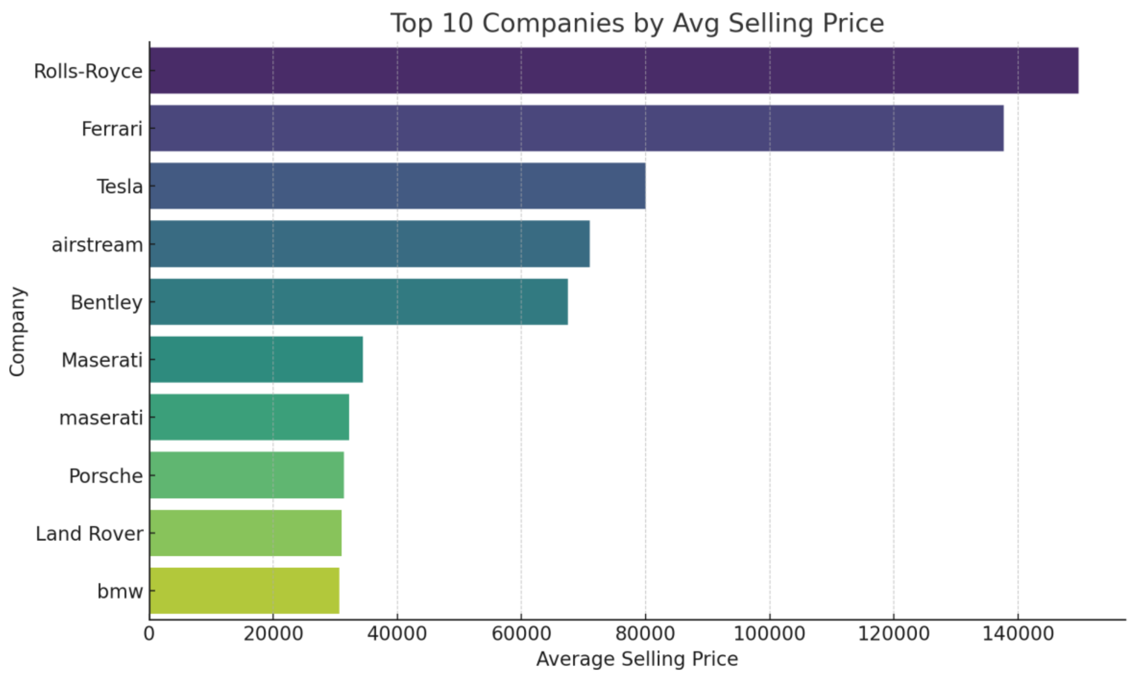


Figure 2: Distribution of Car Grades

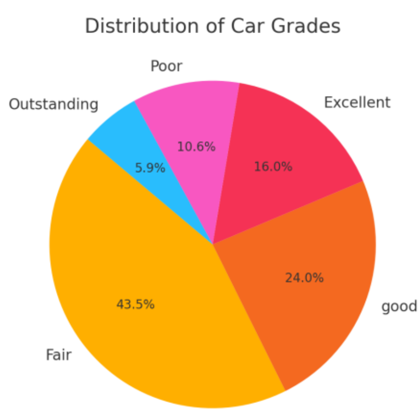
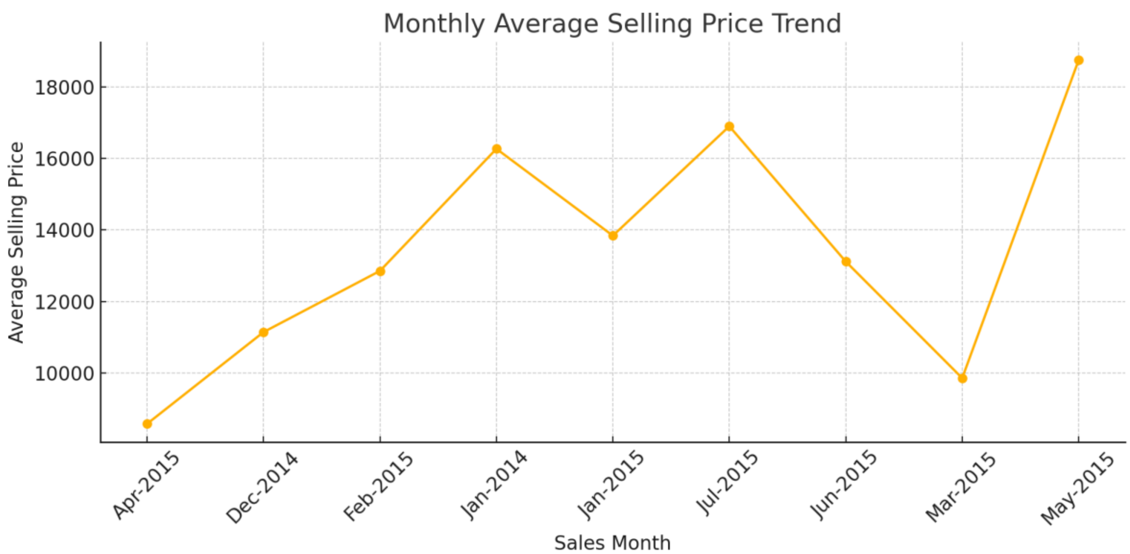
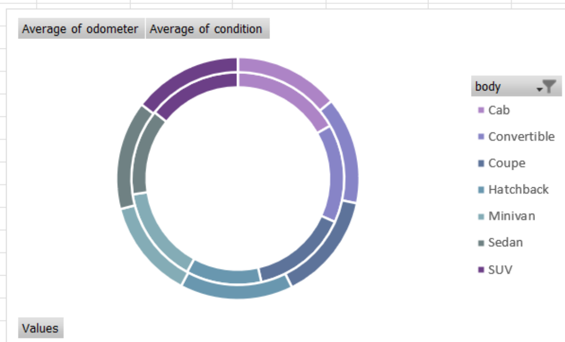


Figure 3: Monthly Average Selling Price Trend



Row Labels	Average of odometer	Average of condition
Cab	96864.92345	30.24217063
Convertible	86721.55494	30.86556169
Coupe	85362.56734	31.07097458
Hatchback	66456.15589	32.54237288
Minivan	85044.78311	28.55742935
Sedan	75846.6084	31.33202621
SUV	82729.44563	31.44771293
Grand Total	**80280.02549**	**31.17217019**



Supplementary Dataset Views

Figure 4: Sample of Raw Dataset

Sample of Raw Dataset

year	company	model	trim	body	transmissi	vin	state	Grade	condition	odomete	color	interior	seller	mmr	sellingpric	day	month	date	year2	Column2	Column3	salesmonth
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg56	PB	Poor	5	16639.0	white	black	kia motors ame	20500	21500	Tue	Dec	16	2014	12:30:00	GMT-0800 (P	Dec-2014
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg56	PB	Poor	5	9393.0	white	beige	kia motors ame	20800	21500	Tue	Dec	16	2014	12:30:00	GMT-0800 (P	Dec-2014
2014	BMW	3 Series	328i SULE	Sedan	automatic	wba3c1c51ek1	PB	Excellent	45	1331.0	gray	black	financial services rema	31900	30000	Thu	Jan	15	2015	04:30:00	GMT-0800 (P	Jan-2015
2015	Volvo	S60	T5	Sedan	automatic	yv1612tb4f13	PB	Excellent	41	14282.0	white	black	volvo na rep/wo	27500	27750	Thu	Jan	29	2015	04:30:00	GMT-0800 (P	Jan-2015
2014	BMW	6 Series Gran	650i	Sedan	automatic	wba6b2c57ed1	PB	Excellent	43	2641.0	gray	black	financial services rema	66000	67000	Thu	Dec	18	2014	12:30:00	GMT-0800 (P	Dec-2014

Figure 5: Sample of Cleaned Dataset

Sample of Cleaned Dataset

year	company	model	trim	body	transmissi	vin	state	Grade	condition	odomete	color	interior	seller	mmr	sellingpric	day	month	date	year2	Column2	Column3	salesmonth
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg56	PB	Poor	5	16639.0	white	black	kia motors ame	20500	21500	Tue	Dec	16	2014	12:30 GMT	-0800 (P	Dec-2014
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg56	PB	Poor	5	9393.0	white	beige	kia motors ame	20800	21500	Tue	Dec	16	2014	12:30 GMT	-0800 (P	Dec-2014
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2014	BMW	6 Series Gran	650i	Sedan	automatic	wba6b2c57ed1	PB	Excellent	43	2641.0	gray	black	financial services rema	66000	67000	Thu	Dec	18	2014	12:30 GMT	-0800 (P	Dec-2014

These charts and visuals helped to represent the key insights obtained from the dataset in a visually appealing and user-friendly manner. They allow users to quickly interpret the results and focus on critical areas for better analysis and reporting.

Chapter 6 – Dashboard

The final dashboard consolidates all key visualizations and KPIs into one dynamic sheet.
It allows users to filter by:

- Brand
- Year
- Month
- Transmission Type
- Grade

The dashboard includes:

- Bar chart for brand comparison
- Line chart for monthly pricing trends
- Pie and donut charts for grade and condition analysis
- Slicer-based filtering for dynamic interaction
- State-level map for regional selling price visualization

This interactive design enables better understanding of car market trends in India through visual storytelling and data-driven exploration.

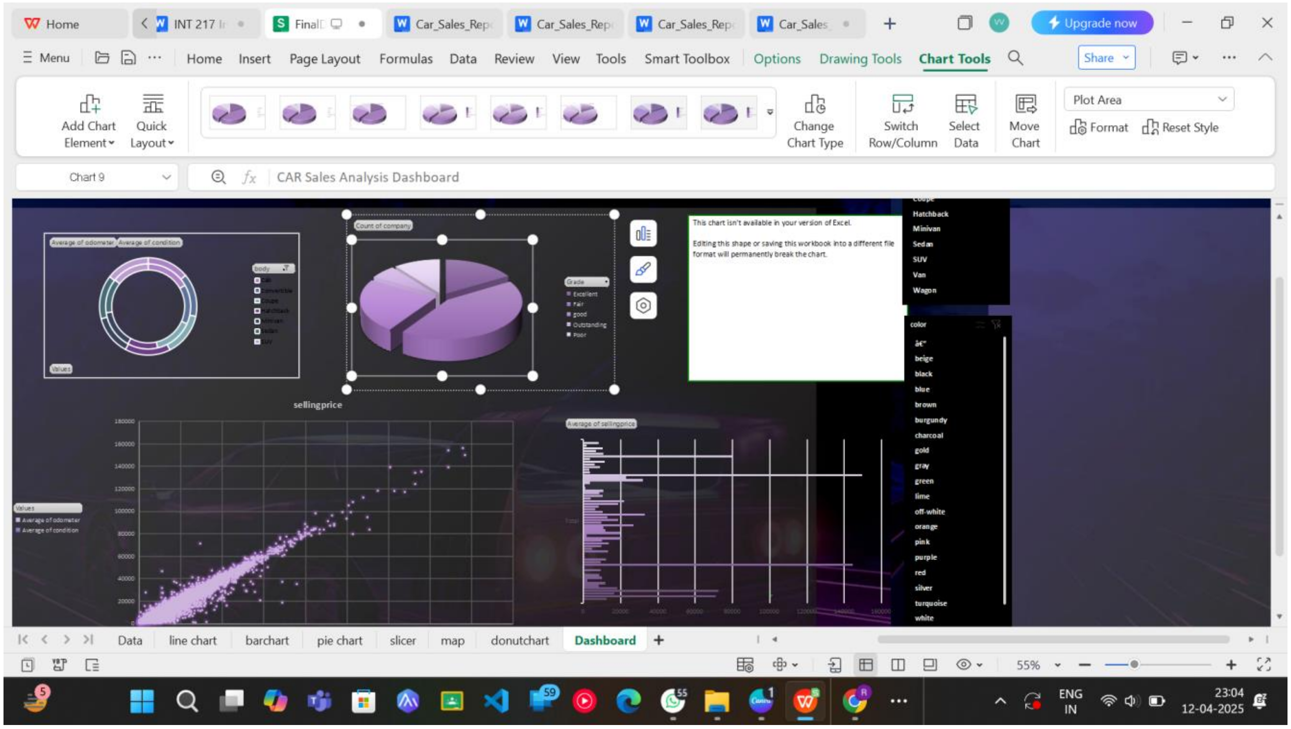

Figure 7: Dashboard Screenshot

Excel Dashboard Snapshot

Unnamed: 3	Unnamed: 30
CAR Sales Analysis Dashboard	nan
nan	

Figure 6: Sample View of the Dashboard

Chapter 7 - Conclusion

This project showcased the importance of structured data analysis and interactive dashboards using Excel. By working with real-world car sales data, we identified trends in pricing, performance, buyer preferences, and regional differences.

Excel tools such as Pivot Tables, Charts, and Slicers enabled the transformation of a raw dataset into a clean, visual, and interactive report. This project enhanced analytical thinking, dashboard creation, and data interpretation skills.

The insights gained through this project are valuable for stakeholders in the automobile sector, including sales analysts, marketing teams, and supply chain planners.

Chapter 8 - Future Scope

The dashboard created in this project offers a foundation for further data-driven work. Future enhancements may include:

- Integration with real-time car listing platforms for dynamic data feeds.
- Expansion to include additional fields like fuel type, insurance status, and buyer demographics.
- Use of Power BI or Tableau for enhanced visualization capabilities and online dashboard sharing.
- Incorporation of machine learning models to forecast prices or detect outliers.

Such advancements can empower dealerships, manufacturers, and analysts to make smarter decisions based on deeper market understanding.

Chapter 9 - References

1. ExcelJet, “Pivot Table Basics,” [Online].

2. D. Paranjape, "Mastering Data Analysis with Excel," Packt Publishing, 2016.
3. Lovely Professional University, "Course Material - INT 217 Introduction to Data Management," School of Computer Science and Engineering, 2025.

Visualizations

LINKDIN: